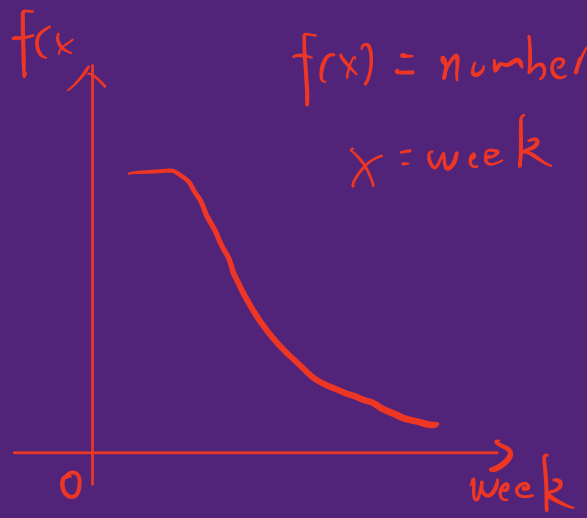

f(x) = number of student

x = week

Start at 8:03

ECON1050 Tutorial 4

Calculus 1: Limits and Differentiation

Math 3404

2 students

Week 5 - Semester 1 2026

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Today's Plan

- Recap: key ideas from Lecture 4
- Tutorial questions: Q1–Q6
- Further exercises: not done in tutorial...

Recap: Limits

What a limit means

$$\boxed{\lim_{x \to a} f(x) = L}$$

as x approaches a, $f(x)$ approaches L

means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a.

constant

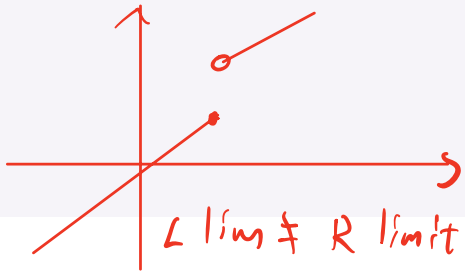
∞

constant : convergence

∞ : divergence

Key ideas

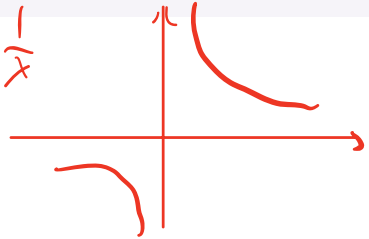
- A limit depends on the values of $f(x)$ near a, not necessarily on the value $f(a)$. $\lim \neq f(a)$
- A two-sided limit takes into account values of x on both sides of a.
- If the left-hand and right-hand limits differ, there is no single limit.
- If the function becomes unbounded near a, the limit does not exist.



L lim ≠ R limit

⇒ limit DNE

$f(x) = \frac{1}{x}$



at $x = 0$

Recap: Univariate differentiation

one input, one variable

Main idea

Univariate differentiation measures the rate of change of one variable in response to a change in another variable. Geometrically, it gives the slope of the tangent to a function at a point.

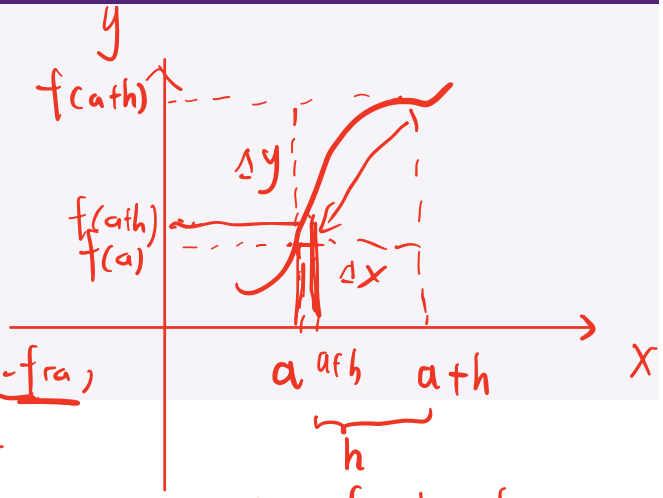
From secant slope to derivative

For a secant line,

$$\frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a)}{h}.$$

The derivative at $x = a$ is obtained by taking the limit as $h \to 0$:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$



$$\frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a)}{h}$$

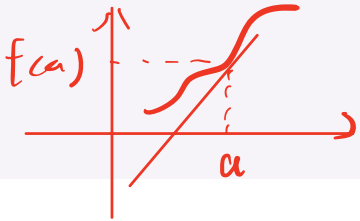
$$\Delta x \to 0 \Leftrightarrow h \to 0 \Rightarrow \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = f'(a)$$

Recap: Tangent line, notation, and differentiability

Tangent line at $x = a$

$y - f(a) = f'(a)(x-a)$

$$y - f(a) = f'(a)(x - a).$$



The slope of the tangent line is the first derivative evaluated at the point.

Common notation

$$\boxed{f'(x),} \qquad y', \qquad \frac{dy}{dx}.$$

Differentiability and continuity

If a function is differentiable at a point, then it must be continuous there. However, continuity alone is not enough to guarantee differentiability.

f(x) is smooth enough : no sudden jump

Recap: First derivatives and rates of change

Increasing and decreasing functions

For an interval I:

- $f'(x) \geq 0$ on I $\Rightarrow$ f is increasing on I.
- $f'(x) > 0$ on I $\Rightarrow$ f is strictly increasing on I.
- $f'(x) \leq 0$ on I $\Rightarrow$ f is decreasing on I.
- $f'(x) < 0$ on I $\Rightarrow$ f is strictly decreasing on I.

Rates of change

The derivative gives the **instantaneous rate of change**. The **relative rate of change** at $x = a$ is

$$\frac{f'(a)}{f(a)}.$$

$$\frac{f'(a)}{f(a)}$$

Recap: Basic derivative rules I

Constants and powers

$$\frac{d}{dx}(c) = 0, \qquad \frac{d}{dx}\big(cf(x)\big) = cf'(x), \qquad \boxed{\frac{d}{dx}(x^n) = nx^{n-1}}.$$

Power function

Sums and differences

If $F(x) = f(x) \pm g(x)$, then

$$F'(x) = f'(x) \pm g'(x).$$

$F(x) = f(x) \pm g(x)$

Recap: Basic derivative rules II

Product rule

If $F(x) = f(x)g(x)$, then

$$F'(x) = f'(x)g(x) + f(x)g'(x).$$

$f' \cdot g + f \cdot g'$

Quotient rule

If $F(x) = \dfrac{f(x)}{g(x)}$ and $g(x) \neq 0$, then

$F(x) = \dfrac{f(x)}{g(x)}$

$$F'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}.$$

$\dfrac{f' \cdot g - f \cdot g'}{g^2}$

Chain rule

If $y = f(g(x))$, then

$y = f(g(x))$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x))g'(x).$$

$y' = f'(g(x)) \cdot g'(x)$

Recap: Special derivatives and second derivatives

Exponential and logarithmic functions

Using the chain rule,

$$\frac{d}{dx}(e^x) = e^x, \qquad \frac{d}{dx}(a^x) = (\ln a)a^x, \qquad \frac{d}{dx}(\ln x) = \frac{1}{x}.$$

$$\frac{d}{dx}(e^{g(x)}) = e^{g(x)}g'(x), \qquad \frac{d}{dx}(\ln(g(x))) = \frac{g'(x)}{g(x)}.$$

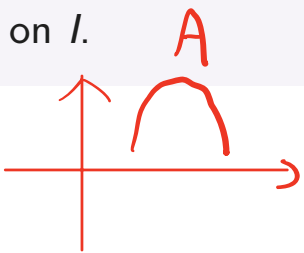
Higher-order derivatives

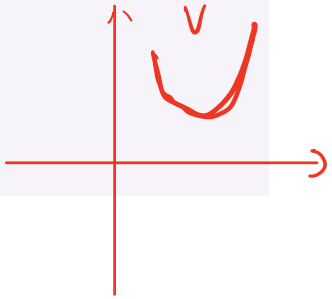
The second derivative $f''(x)$ describes how the first derivative changes.

- $f''(x) \geq 0$ on I $\Rightarrow$ f is convex on I.
- $f''(x) \leq 0$ on I $\Rightarrow$ f is concave on I.

Convex iff $f''(x) \geq 0$

Concave iff $f''(x) \leq 0$





Tutorial Q1

Question 1

Differentiate:

(a) $y = \frac{1}{3}x^9$

(b) $y = 3x^7 + 8$

(c) $y = 5x^3 + 3x - 1$

Q1 c

Q2 bd

7 min

(c) $y = 5x^3 + 3x - 1$, $f(x) = 5x^3$, $g(x) = 3x - 1$

$\Rightarrow f'(x) = 5 \cdot 3x^2 = 15x^2$

$g'(x) = 3$

$y = f(x) + g(x)$

$y' = f'(x) + g'(x)$

$= 15x^2 + 3$ ✓

Tutorial Q1: Solution

Knowledge used

- Power rule
- Derivative of a constant
- Sum rule

Solution

(a)

$$y' = \frac{1}{3}(9)x^8 = 3x^8.$$

(b)

$$y' = 3(7)x^6 + 0 = 21x^6.$$

(c)

$$y' = 5(3)x^2 + 3(1)x^0 - 0 = 15x^2 + 3.$$

Tutorial Q2

Question 2

Differentiate:

(a) $y = (x^3 - 1)(x^2 + 2)$

(b) $g(t) = \dfrac{2t+1}{t^2+3}$

(c) $y = 5x^2 + \dfrac{2x}{x^2+3}$

(d) $y = (3x - 1)(2x^2 + 3)$ product Rule

Q2 (b) $g(t) = \dfrac{2t+1}{t^2+3} \Rightarrow f(t) = 2t+1 \Rightarrow f'(t) = 2$

$h(t) = t^2+3 \Rightarrow h'(t) = 2t$

Quotient rule $g'(t) = \dfrac{f' \cdot h - f \cdot h'}{h^2} = \dfrac{2 \cdot (t^2+3) - (2t+1) \cdot (2t)}{(t^2+3)^2}$

$= \dfrac{2t^2+6-4t^2-2t}{(t^2+3)^2} = \dfrac{-2t^2-2t+6}{(t^2+3)^2}$ ✓

(d) $y = (3x-1)(2x^2+3)$

$= f(x) \cdot g(x)$

$f(x) = 3x-1 \Rightarrow f' = 3$

$g(x) = 2x^2+3 \Rightarrow g' = 4x$

$y' = f' \cdot g + f \cdot g' = 3 \cdot (2x^2+3) + (3x-1) \cdot 4x$

$= 6x^2+9+12x^2-4x = 18x^2-4x+9$ ✓

Tutorial Q2: Solution I

Knowledge used

- Product rule
- Quotient rule
- Power rule and sum rule

Solution

(a)

$$y' = (3x^2)(x^2+2) + (x^3-1)(2x) = 3x^4 + 6x^2 + 2x^4 - 2x = 5x^4 + 6x^2 - 2x.$$

(b)

$$g'(t) = \frac{2(t^2+3) - (2t+1)(2t)}{(t^2+3)^2} = \frac{-2t^2 - 2t + 6}{(t^2+3)^2}.$$

Tutorial Q2: Solution II

Knowledge used

- Sum rule plus quotient rule
- Product rule

Solution

(c)

$$y' = 10x + \frac{2(x^2+3) - 2x(2x)}{(x^2+3)^2} = 10x + \frac{6-2x^2}{(x^2+3)^2}.$$

(d)

$$y' = 3(2x^2+3) + (3x-1)(4x) = 6x^2 + 9 + 12x^2 - 4x = 18x^2 - 4x + 9.$$

Tutorial Q3

Chain Rule

Question 3

Differentiate:

(a) $y = 10(5 - x^2)^2$

(b) $y = (3x^2 + 6x + 6)^{1/2}$

(c) $y = \sqrt{\frac{1}{x} - 1}$

Power rule

(a) $y = 10(5 - x^2)^2$, $g(x) = 5 - x^2 \Rightarrow g'(x) = -2x$

$f(x) = 10 \cdot x^2 \Rightarrow f'(x) = 20x$

$y = f(g(x))$

$y' = f'(g(x)) \cdot g'(x)$

$y' = 20 \cdot (5 - x^2) \cdot (-2x) = -40x(5 - x^2) = -200x + 40x^3$ ✓

(c) $y = \sqrt{\frac{1}{x} - 1} = \left(\frac{1}{x} - 1\right)^{\frac{1}{2}}$

$g(x) = \frac{1}{x} - 1 = (x)^{-1} - 1 \Rightarrow g'(x) = -(x)^{-2}$

$f(x) = x^{\frac{1}{2}} \Rightarrow f'(x) = \frac{1}{2} x^{-\frac{1}{2}}$

$y = f(g(x))$

$y' = f'(g(x)) \cdot g'(x)$

$= \frac{1}{2} \cdot (g(x))^{-\frac{1}{2}} \cdot (-(x)^{-2}) = \frac{1}{2}(x^{-1} - 1)^{-\frac{1}{2}}(-x^{-2}) = \frac{1}{2} \frac{1}{\left(\frac{1}{x} - 1\right)^{\frac{1}{2}}} \cdot \frac{-1}{x^2} = -\frac{1}{2} \cdot \frac{1}{x^2\sqrt{\frac{1}{x} - 1}}$

$= -\frac{1}{2} \cdot \frac{1}{\sqrt{x^4\left(\frac{1}{x} - 1\right)}} = -\frac{1}{2} \frac{1}{\sqrt{x^3 - x^4}}$

Tutorial Q3: Solution I

Knowledge used

- Chain rule
- Power rule
- Identify the inner and outer functions

Solution

(a) Let $u = 5 - x^2$. Then $y = 10u^2$, so

$$\frac{dy}{dx} = 20u(-2x) = 20(5 - x^2)(-2x) = 40x^3 - 200x.$$

(b) Let $u = 3x^2 + 6x + 6$. Then $y = u^{1/2}$, so

$$\frac{dy}{dx} = \frac{1}{2}u^{-1/2}(6x + 6) = \frac{3x + 3}{(3x^2 + 6x + 6)^{1/2}}.$$

Tutorial Q3: Solution II

Knowledge used

- Rewrite the square root as a power
- Use the chain rule with an inner reciprocal expression

Solution

Write

$$y = \left(\frac{1}{x} - 1\right)^{1/2}.$$

Let $u = \frac{1}{x} - 1$. Then $u' = -x^{-2}$, so

$$\frac{dy}{dx} = \frac{1}{2}u^{-1/2}(-x^{-2}) = \boxed{-\frac{1}{2\sqrt{x^3 - x^4}}}.$$

Tutorial Q4

Question 4

Find the first and second derivatives.

(a) $z = 120t - \frac{1}{3}t^3$

(b) $y = \sqrt{x} = x^{1/2}$ for $x > 0$

(c) $g(t) = \frac{t^2}{t-1}$, $t \neq 1$

(a) $z = 120t - \frac{1}{3}t^3$

$z' = 120 - \frac{1}{3}\cdot 3\cdot t^2 = 120 - t^2$ ✓

$z'' = 0 - 2t = -2t$

(c) $g(t) = \frac{t^2}{t-1}$

Quotient rule $\begin{cases} f(t) = t^2 \Rightarrow f' = 2t \\ h(t) = t-1 \Rightarrow h' = 1 \end{cases}$

$g'(t) = \frac{f'\cdot h - f\cdot h'}{h^2} = \frac{2t(t-1) - t^2\cdot 1}{(t-1)^2} = \frac{2t^2 - 2t - t^2}{(t-1)^2} = \boxed{\frac{t^2-2t}{(t-1)^2}}$

Quotient rule

$g''(t) = \frac{(2t-2)(t-1)^2 - (t^2-2t)\cdot 2(t-1)}{(t-1)^4} = \frac{(2t-2)(t^2-2t+1) - 2(t^2-2t)(t-1)}{(t-1)^4} = \dots$

Simply algebra

check verify $\Rightarrow$ $= \frac{2}{(t-1)^3}$

Tutorial Q4: Solution I

Knowledge used

- Power rule
- Higher-order derivatives

Solution

(a)

$$z' = 120 - \frac{1}{3}(3)t^2 = 120 - t^2, \qquad z'' = -2t.$$

(b)

$$y' = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}, \qquad y'' = -\frac{1}{4}x^{-3/2} = -\frac{1}{4\sqrt{x^3}}.$$

Tutorial Q4: Solution II

Knowledge used

- Quotient rule
- Differentiate again to obtain the second derivative

Solution

Using the quotient rule,

$$g'(t) = \frac{2t(t-1) - t^2}{(t-1)^2} = \frac{t^2 - 2t}{(t-1)^2}.$$

Differentiating again,

$$g''(t) = \frac{2}{(t-1)^3}.$$

Tutorial Q5

Question 5

(a) Determine where the function is $\boxed{\text{strictly increasing:}}$ $\Leftrightarrow$ $y' > 0$

$$y = (\ln x)^2 - 4.$$

(b) For

$$f(x) = 2x^3 - 12x^2 + 5,$$

determine where f is increasing, concave, and convex.

$f' \geq 0$ $\Updownarrow$

chain rule $\Downarrow$

f'' $f'' \leq 0$ $f'' \geq 0$

(b) $f'(x) = 2 \cdot 3x^2 - 12 \cdot 2x$
$= 6x^2 - 24x$

$f''(x) = 6 \cdot 2 \cdot x - 24$
$= 12x - 24$

Increasing $\Leftrightarrow$ $f'(x) = 6x^2 - 24x \geq 0$
$= 6x(x-4) \geq 0$
$\Rightarrow (-\infty, 0] \cup [4, \infty)$

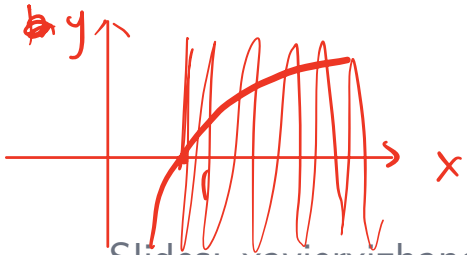
		0		4	
$6x$	$-$		$+$		$+$
$x-4$	$-$		$-$		$+$
	✓			✓	

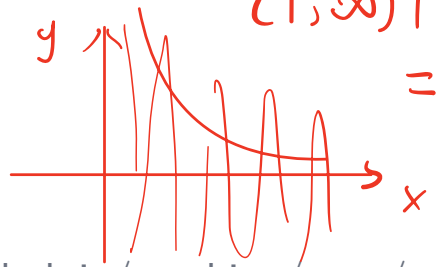
(a) $y = (\ln x)^2 - 4$

$(\ln x)' = \frac{1}{x}$

$y' = 2 \cdot (\ln x) \cdot \frac{1}{x} > 0$

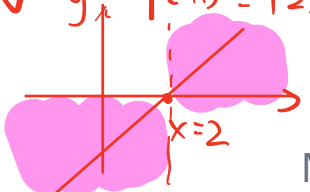
$(1, \infty) \cap (0, \infty)$
$= (1, \infty)$ ✓





Concave and convex $\Rightarrow f''(x) \begin{cases} \leq 0 : \text{concave} \\ \geq 0 : \text{convex} \end{cases}$

$f''(x) = 12x - 24$, $f''(x) = 0 \Rightarrow x = 2$



concave: $(-\infty, 2]$

convex: $[2, \infty)$

Tutorial Q5: Solution I

Knowledge used

- Use the first derivative to determine monotonicity
- Analyse the sign of $\ln x$ on its domain

Solution

$$y' = 2(\ln x)\frac{1}{x} = \frac{2\ln x}{x}.$$

For $0 < x < 1$, we have $\ln x < 0$. For $x > 1$, we have $\ln x > 0$. Hence

$$y' > 0 \text{ for } x > 1.$$

Therefore the function is strictly increasing for

$$x > 1.$$

Tutorial Q5: Solution II

Knowledge used

- Use the first derivative for increasing/decreasing intervals
- Use the second derivative for concavity/convexity

Solution

$$f'(x) = 6x^2 - 24x = 6x(x-4), \qquad f''(x) = 12x - 24 = 12(x-2).$$

Therefore:

- f is increasing on
$$(-\infty, 0] \cup [4, \infty).$$
- f is concave on
$$(-\infty, 2].$$
- f is convex on
$$[2, \infty).$$

Tutorial Q6

Question 6

Find the equation of the tangent to

$$y = x^{1/2} - x^2$$

at $x = 4$.

3 习2 min

General form: $y - f(a) = f'(a)(x-a)$

$f(x)$ is the function,

$f(x) = x^{\frac{1}{2}} - x^2$, $a = 4$

① $f(4)$: $f(4) = 4^{\frac{1}{2}} - 4^2 = 2 - 16 = \underline{-14}$

② $f'(4)$: $f'(x) = \frac{1}{2}x^{-\frac{1}{2}} - 2x = \frac{1}{2}\cdot\frac{1}{\sqrt{x}} - 2x \Rightarrow f'(4) = \frac{1}{2}\cdot\frac{1}{\sqrt{4}} - 2\cdot 4 = \frac{1}{4} - 8 = \underline{-\frac{31}{4}}$

$\Rightarrow y - (-14) = -\frac{31}{4}(x-4) \Leftrightarrow y + 14 = -\frac{31}{4}(x-4)$

simplify or done

Tutorial Q6: Solution

Knowledge used

- Differentiate to find the slope at the given point
- Use the slope-point formula $y - y(a) = y'(a)(x - a)$

Solution

$$y' = \frac{1}{2}x^{-1/2} - 2x = \frac{1}{2\sqrt{x}} - 2x.$$

At $x = 4$,

$$y'(4) = \frac{1}{4} - 8 = -\frac{31}{4}, \qquad y(4) = \sqrt{4} - 4^2 = -14.$$

So the tangent line is

$$y - (-14) = -\frac{31}{4}(x - 4),$$

which simplifies to

$$y = -\frac{31}{4}x + 17.$$

Further exercise: Q7(e)

Extra Practice Question 7(e)

Differentiate:

$$y = x\ln(x^2 + 1).$$

Knowledge used

- Product rule
- Chain rule for $\ln(g(x))$

Solution

Using the product rule and the chain rule,

$$y' = \ln(x^2 + 1) + x \cdot \frac{2x}{x^2 + 1} = \ln(x^2 + 1) + \frac{2x^2}{x^2 + 1}.$$

Further exercise: Q8

Extra Practice Question 8

For

$$y = \frac{x^2 - x^3}{x+3}, \qquad x \neq -3,$$

find the equation of the tangent at $x = 1$.

Knowledge used

- Quotient rule
- Tangent line formula at a specified point

Solution

$$y' = \frac{(2x - 3x^2)(x+3) - (x^2 - x^3)}{(x+3)^2} = \frac{-2x^3 - 8x^2 + 6x}{(x+3)^2}.$$

At $x = 1$, $y'(1) = -\frac{1}{4}$ and $y(1) = 0$, so

$$y = -\frac{1}{4}x + \frac{1}{4}.$$

Further exercise: Q9

Extra Practice Question 9

Suppose $A(x)$ is the cost in dollars of building with a floor size of x square metres, and

$$A'(100) = 250.$$

Interpret this result.

Knowledge used

- The first derivative is a marginal quantity
- In economics, the derivative measures the change from one more unit of the independent variable

Solution

$A'(x)$ is the **marginal cost** of increasing floor area. Hence

$$A'(100) = 250$$

means that when the floor size is 100 square metres, the cost of increasing the floor area from 100 to 101 square metres is approximately

$$\$250.$$

Further exercise: Q10(g)

Extra Practice Question 10(g)

Differentiate:

$$y = a^{5x}, \qquad a > 0.$$

Knowledge used

- Rewrite a general exponential using e
- Apply the chain rule together with $\frac{d}{dx}(a^x) = (\ln a)a^x$

Solution

Write $y = a^{5x} = e^{5(\ln a)x}$. Then

$$\frac{dy}{dx} = 5(\ln a)e^{5(\ln a)x} = 5(\ln a)a^{5x}.$$

Key takeaways

- A limit is determined by values near a point, not necessarily by the function value at the point itself.
- The derivative measures instantaneous change and gives the slope of the tangent line.
- Use the right differentiation rule: power, sum, product, quotient, or chain rule.
- First derivatives help with monotonicity; second derivatives help with concavity and convexity.
- In economics, derivatives often have marginal interpretations.